Assignment 2

SUBJECT- SAPM F412

TOPIC- Technical Analysis of the Stock

GROUP- 06

SUBMITTED BY

Name	ID NO.
Rishikant Kumar	2023B3A10829G
Abhishek Bs	2023B3A30834G
Priyanshu Jain	2023B3A70835G
Hardik Agarwal	2023B3A70838G
Aarush Gupta	2023B3A70839G
Sajal Mutneja	2023B3AD0841G
Aditya Agarwal	2023B3A40831G
Arya Rajeev Shah	2023B3A70842G

SAPM Assignment on Technical Analysis		
Fundamentals	**Aspect**	**Details**
	Economic Environment	The Indian economy is expected to grow steadily, providing a conducive environment for banking growth.
	Loan Growth	Retail loan book growth is projected at 20% year-on-year, driven by increased consumer demand.
	Deposit Mobilization	Retail term deposits have grown by 17% year-on-year as customers seek safe investment avenues.
	Digital Transformation	Continued investment in digital capabilities is anticipated to enhance customer engagement further.
	Corporate Banking Outlook	A 7% growth in corporate loans is expected as businesses recover post-pandemic.
	Risk Management	Enhanced risk management frameworks are being implemented to mitigate potential economic shocks.
	Sustainability Initiatives	Focusing on sustainable finance will continue to emphasize green projects and social impact lending.
Trends Overview	**Primary Trend (2 June 2023):** Overall upward trajectory signifying strong long-term growth momentum. **Secondary Trends:** 13 June to 16 August: A temporary downward correction within the primary uptrend. 3 October to 1 November: Another shorter correction before resuming growth. **Minor Trends:** Notable fluctuations (e.g., June and October) represent short-term trading opportunities, driven by market sentiment shifts.	
Candlestick Patterns	The analysis of Axis Bank's stock combines key trends, candlestick patterns, and moving average signals to track market movements throughout 2023. The Bullish Engulfing on February 27 marked a strong reversal from a downtrend, sparking an upward move later tested with a Doji on May 24, indicating market indecision. This led to continued gains until a Bearish Engulfing appeared on July 27, signaling a short-term bearish reversal. After consolidating, the stock displayed a Morning Star on November 13, heralding a robust bullish trend that peaked with a Shooting Star on December 6, indicating potential overbought conditions and a looming correction. The Moving Average Double crossover strategy guided key actions, including buying on April 11 at ₹853.3 and November 12 at ₹1029.55, and selling during downtrends on July 27, October 3, and December 28, securing profits. These movements align with the broader primary uptrend seen from June 2, supported by secondary trends like the decline from June 13 to August 16 and subsequent rallies from October 3 to November 1. This comprehensive strategy,	

	reinforced by technical and fundamental analysis, delivered a portfolio growth of 30.34% by year-end.					
Leading and Lagging Indicators	**Date**	**Action**	**Price (₹)**	**Shares Held**	**Portfolio Value (₹)**	**Reason for Action**
	23-Jan	**Sell**	N/A	0	Not invested	Moving Average Double (14 crosses below 21) and RSI indicates overbought conditions.
	11-Apr	**Buy**	853.3	1,172	₹ 9,99,887.60	Moving Average Double (14 crosses above 21) and RSI signals oversold or reversal potential.
	27-Jul	**Sell**	973.25	0	₹ 11,41,224.50	Moving Average Double signals weakening trend (14 close to crossing below 21), and RSI declines.
	24-Aug	**Buy**	981.35	1,163	₹ 11,40,961.60	Moving Average Double crossover (14 above 21), supported by RSI recovery from lower levels.
	03-Oct	**Sell**	1,041.05	0	₹ 12,11,944.15	RSI near overbought; price nearing resistance; Moving Averages remain flat, signaling stagnation.

	12-Nov	**Buy**	1,029.95	1,176	₹ 12,11,352.20	Moving Average Double (14 crosses above 21); RSI recovery suggests bullish momentum.
	28-Dec	**Sell**	1,107.85	0	₹ 13,03,419.60	RSI overbought, price peaking, and divergence visible between price and Moving Averages.

Fundamental

Axis Bank, one of India's leading private sector banks, has established a strong reputation over its 30 years of operations by focusing on customer-centric banking solutions and innovation. The bank's ethos, encapsulated in the phrase "dil se open," emphasizes openness and accessibility, which are central to its strategy. Axis Bank offers a wide range of financial services, including retail and wholesale banking, wealth management, and digital banking solutions. The bank has made significant investments in technology to enhance customer experience and operational efficiency, exemplified by its mobile banking app, which facilitates seamless transactions for millions of users.

Business Fundamentals

Business Model and Segments

Axis Bank delivers a broad spectrum of financial services, catering to individuals, businesses, and corporations. Its operations are organized across key business segments:

1. **Retail Banking**
   This segment serves individual consumers and small businesses. The offerings include savings and current accounts, home and personal loans, auto loans, credit cards, and wealth management services. Retail banking is a major growth engine for the bank, with

a vast customer base supported by an extensive branch and ATM network.

2. **Wholesale Banking**
   Focused on corporate clients and mid-sized businesses, this division provides customized financial solutions such as working capital loans, project finance, trade services, and treasury products. The bank's ability to service the complex financial needs of businesses has helped it become a trusted partner for corporate India.

3. **Treasury Operations**
   This unit manages the bank's investment portfolio and liquidity. It is also involved in foreign exchange, derivatives, and other risk management services. Efficient treasury operations are critical for maintaining financial stability and optimizing returns on surplus funds.

4. **Digital and Technology-Driven Services**
   A key strength of Axis Bank is its investment in digital innovation. The bank has embraced modern technologies to enhance customer experience, ensure operational efficiency, and expand reach. Its mobile banking app and internet banking platforms offer convenient, secure, and feature-rich services to millions of customers.

Strategic Focus and Differentiation

Axis Bank's strategic focus revolves around digital transformation, service excellence, and sustainable growth. The bank has been proactive in deploying emerging technologies such as artificial intelligence, big data analytics, and machine learning to personalize banking experiences and improve decision-making.

Customer engagement is driven by a combination of responsive service, competitive products, and digital convenience. Moreover, the bank has been placing a growing emphasis on environmental, social, and governance (ESG) goals, integrating sustainability into its long-term business vision.

Performance and Market Position

Axis Bank is among India's top private sector banks in terms of assets, customer base, and branch network. Its diversified revenue streams and prudent risk management practices have enabled it to navigate economic cycles effectively. The bank maintains a healthy capital adequacy ratio and has steadily improved its asset quality through robust credit underwriting and proactive risk controls.

Near-Term Fundamentals from Management Discussion & Analysis

Aspect	Details
Economic Environment	The Indian economy is expected to grow steadily, providing a conducive environment for banking growth.
Loan Growth	Retail loan book growth is projected at 20% year-on-year, driven by increased consumer demand.
Deposit Mobilization	Retail term deposits have grown by 17% year-on-year as customers seek safe investment avenues.

Digital Transformation	Continued investment in digital capabilities is anticipated to enhance customer engagement further.
Corporate Banking Outlook	A 7% growth in corporate loans is expected as businesses recover post-pandemic.
Risk Management	Enhanced risk management frameworks are being implemented to mitigate potential economic shocks.
Sustainability Initiatives	Focusing on sustainable finance will continue to emphasize green projects and social impact lending.

Conclusion:

Axis Bank's strategic focus on innovation, customer service, and sustainability positions it well for future growth. The bank's proactive approach to adapting to market conditions and embracing digital transformation will likely yield positive outcomes in the near term. As it continues to expand its reach across urban and rural areas while enhancing its service offerings through technology, Axis Bank aims to solidify its status as a preferred banking partner in India.

Through these efforts outlined in the MD&A section of the Annual Report, Axis Bank demonstrates a clear roadmap for achieving sustained growth while contributing positively to the economy and society.

Analysis of Candlestick Patterns:

Date	Action	Price (₹)	Shares Held	Portfolio Value (₹)	Reason for Action
23-Jan	Sell	N/A	0	Not invested	Moving Average Double (14 crosses below 21) and RSI indicates overbought conditions.
11-Apr	Buy	853.3	1,172	₹ 9,99,887.60	Moving Average Double (14 crosses above 21) and RSI signals oversold or reversal potential.
27-Jul	Sell	973.25	0	₹ 11,41,224.50	Moving Average Double signals weakening trend (14 close to crossing below 21), and RSI declines.
24-Aug	Buy	981.35	1,163	₹ 11,40,961.60	Moving Average Double crossover (14 above 21), supported by RSI recovery from lower levels.
03-Oct	Sell	1,041.05	0	₹ 12,11,944.15	RSI near overbought; price nearing resistance; Moving Averages remain flat, signaling stagnation.
12-Nov	Buy	1,029.95	1,176	₹ 12,11,352.20	Moving Average Double (14 crosses above 21); RSI recovery suggests bullish momentum.
28-Dec	Sell	1,107.85	0	₹ 13,03,419.60	RSI overbought, price peaking, and divergence visible between price and Moving Averages.

Portfolio Growth Summary
Starting Value: **₹1,000,000** Ending Value: **₹1,303,419.6** Total Profit: **₹303,419.6 (+30.34%)**
Signal Effectiveness Evaluation
11 April Buy: Signal was accurate. Price increased significantly after the buy signal.
27 July Sell: Accurate. Price declined after this point, preventing potential losses.
24 August Buy: Accurate. Price increased after the buy.
3 October Sell: Accurate. Price peaked soon after, and the decision avoided holding during a stagnation phase.
12 November Buy: Signal was valid but less impactful. While there was a mild increase in price, the trade did not yield significant growth.
28 December Sell: Accurate. Price was at a high; selling locked in significant profits.

Fundamental Analysis:

The analysis evaluates Axis Bank's stock valuation using multiple methods:

- PE Ratio
- PEG Ratio
- Market to Book Value
- Discounted Cash Flow (DCF) Method

Key Inputs

Metric	Value
Market Price per Share (MPS)	₹1047.2
Earnings per Share (EPS)	₹80.67
Retention Ratio	0.988
Return on Equity (ROE)	0.186
Book Value per Share	₹486.74
Growth Rate	0.183768
Discount Rate (WACC)	0.1729
Terminal Growth Rate	0.05
Outstanding Shares	3,086,570,375

Valuation Ratios & Analysis

1. PE Ratio

- **Actual:** 12.98
- **Industry Avg (Private Sector):** 16.64
- **Estimated Price:** ₹1342.35
- **Remark: Underpriced**

2. PEG Ratio

- **Actual:** 0.706
- **Industry Benchmark:** 1
- **Estimated Price:** ₹1482.46
- **Remark: Underpriced**

3. Market to Book Value

- **Actual:** 2.15
- **Industry Average:** 2.28
- **Estimated Price:** ₹1109.77
- **Remark: Underpriced**

The above valuation indicators suggest the stock is trading below its fair value.

Discounted Cash Flow (DCF) Method

Forecasted Free Cash Flow (FCF) (₹ in Crores)

Year	Forecasted FCF	Discounted FCF
1	28,000	23,872.45
2	31,000	22,534.07
3	34,000	21,071.52
4	37,000	19,550.50
5	40,000	18,020.01

Terminal Value (Year 5): ₹341,741.25 Cr
Discounted Terminal Value: ₹153,954.53 Cr
Firm Intrinsic Value (Sum of discounted FCFs): ₹259,003.09 Cr
Intrinsic Value per Share: ₹839.13

Current MPS: ₹1047.2
DCF Remark: Underpriced

Note: Despite the DCF-derived intrinsic value (₹839.13) being lower than the current MPS (₹1047.2), other valuation metrics indicate a significantly higher potential value, suggesting a mixed but cautiously optimistic outlook.

Conclusion

Fundamental analysis through PE ratio, PEG ratio, Market to Book Value of Equity and DCF method converge to the conclusion that Axis Bank as of March 2024 was underpriced.

The majority of valuation methods suggest that **Axis Bank is undervalued**, with estimated fair values ranging from ₹1109 to ₹1482 per share. The DCF method, while slightly more conservative, still supports the underpricing narrative when accounting for long-term growth potential.

Investor Takeaway

Axis Bank appears to be a strong fundamental stock with:

Solid earnings

High retention and reinvestment (Retention Ratio: 98.8%)

ROE of 18.6%

Attractive valuation ratios across the board

Investors may consider this an opportunity for medium to long-term growth, especially if the market corrects to align with the intrinsic value estimated by PE, PEG, and market/book ratios.

Final Conclusion

The technical and fundamental analysis of Axis Bank demonstrates strong potential for medium to long-term investment.

Through candlestick pattern recognition and the use of moving average and RSI indicators, timely trades were executed to generate a portfolio return of 30.34%—a performance reflecting both strategic precision and effective signal interpretation.

On the fundamental side, various valuation metrics including PE, PEG, Market-to-Book, and DCF all point toward Axis Bank being undervalued relative to intrinsic worth. While the DCF approach yielded a slightly lower valuation than the market price, the other metrics—especially PE and PEG—highlight substantial upside potential.

The high retention ratio and ROE further validate the bank's solid financial position and reinvestment strength.

Given the synergy between technical indicators and sound fundamentals, Axis Bank presents itself as a robust investment candidate.

The trading strategy used not only minimized downside exposure but capitalized on upward movements, proving reliable with minimal false signals.

Investors seeking quality in the banking sector can view Axis Bank as a fundamentally sound and technically timed opportunity.